STA2002 Probability and Statistics II
Assignment 5

Due on 11:59pm, 2023/11/26

1. Suppose X is a discrete random variable. Let 50 observations give $x = 0, 1, 2$, and 3 respectively 15, 20, 10, and 5 times.

   (a) If we use a chi-square goodness-of-fit statistic to test the null hypothesis that the distribution of X is bin$(3, 0.5)$, calculate the value of the test statistic and state your conclusion, using an $\alpha = 0.05$ significance level.

   (b) Now, suppose we want to test the null hypothesis that the distribution of X is bin$(3, p)$ for some $0 < p < 1$. Give an estimate $\hat{p}$ for p based on the observations.

   (c) Calculate the value of the test statistic for (b) and state your conclusion, using an $\alpha = 0.05$.

   (a) Use Table II with $n = 3, p = 0.5$, the hypothesis probabilities and respective expected values are as follows

	X=0	X=1	X=2	X=3
Probability	0.125	0.375	0.375	0.125
Expected	6.25	18.75	18.75	6.25
Observed	15	20	10	5

$$q = \frac{(15-6.25)^2}{6.25} + \frac{(20-18.75)^2}{18.75} + \frac{(10-18.75)^2}{18.75} + \frac{(5-6.25)^2}{6.25} = 16.67 > 7.815 = \chi^2_{0.05}(4-1)$$

   Reject the null hypothesis that X is bin$(3, 0.5)$ at $\alpha = 0.05$ significance level.

   (b) Out of $50 \times 3 = 150$ trails, the total number of "successes" is $0 \times 15 + 1 \times 20 + 2 \times 10 + 3 \times 5 = 55$. By MLE, $\hat{p} = \frac{55}{150} = 0.367$

   (c) with $\hat{p} = 0.367$,

	X=0	X=1	X=2	X=3
Probability	0.254	0.441	0.256	0.049
Expected	12.68	22.06	12.79	2.47
Observed	15	20	10	5

$$q = \frac{(15-12.68)^2}{12.68} + \frac{(20-22.06)^2}{22.06} + \frac{(15-15.26)^2}{15.26} = 0.621 < 3.841 = \chi^2_{0.05}(3-1-1)$$

   Do not reject the hypothesis that X is bin$(3, p)$ at $\alpha = 0.05$ significance level.

2. Suppose that in 200 passages of one author's works, the word "also" showed the following frequency distribution

Number of "also" (j)	0	1	2	3	4	5	6	7	Total
Frequency of passages (N_j)	22	53	58	39	20	5	2	1	200

Is it reasonable to assume that the distribution of the number of times the author uses "also" in a passage is Poisson (the approximate significance level is taken to be 0.05)?

Hint: We want to test

$$H_0\text{: The distribution is Poisson}(\lambda)$$

against

$$H_1\text{: The distribution is not Poisson}(\lambda).$$

with MLE of λ is $\hat{\lambda} = \bar{x}$.

Solution: Since $n = 200$, the MLE of λ is

$$\begin{aligned}\hat{\lambda} =& \bar{x} = 0 \times \frac{22}{200} + 1 \times \frac{53}{200} + 2 \times \frac{58}{200} + 3 \times \frac{39}{200} \\ &+ 4 \times \frac{20}{200} + 5 \times \frac{5}{200} + 6 \times \frac{2}{200} + 7 \times \frac{1}{200} \\ =& 2.05\end{aligned}$$

We calculate $\{p_{j0}\}_{j=0}^{7}$ according to their MLEs:

$$p_{j0} = p_{j0}(\lambda) = \frac{\lambda^j}{j!}\mathrm{e}^{-\lambda}, \quad j = 0, 1, \ldots, 6, \quad p_{7,0} = 1 - \sum_{j=0}^{6} p_{j0}$$

$$\hat{p}_{j0} = p_{j0}(\hat{\lambda}) = \frac{\hat{\lambda}^j}{j!}\mathrm{e}^{-\hat{\lambda}}, \quad j = 0, 1, \ldots, 6, \quad \hat{p}_{7,0} = 1 - \sum_{j=0}^{6} \hat{p}_{j0}$$

and obtain

j	0	1	2	3	4	5	6	$7(\geqslant 7)$
N_j	22	53	58	39	20	5	2	1
$\hat{p}_{j0}$	0.1287	0.2639	0.2705	0.1848	0.0947	0.0388	0.0132	0.0054
$n\hat{p}_{j0}$	25.747	52.781	54.101	36.969	18.946	7.7681	2.6541	1.0800

If the expected frequencies in some classes are less than 5, they should be combined with an adjacent class. Therefore, we combine the last 3 classes. Since $m = 6$, we have

$$\hat{Q}_{200} = \sum_{j=0}^{5} \frac{(N_j - n\hat{p}_{j0})^2}{n\hat{p}_{j0}} = 2.052 < \chi^2(0.05, 6 - 1 - 1) = 9.488$$

Thus, we cannot reject H_0 at the 0.05 approximate significance level.
Rmk: It's OK if you do not combine the last three classes and do the test with 8 categories.

3. Each of two comparable classes of 15 students responded to two different methods of instructions, giving the following scores on a standardized test:

Class I :	91	42	39	62	55	82	67	44	51	77	61	52	76	41	59
Class II :	80	71	55	67	61	93	49	78	57	88	79	81	63	51	75

(a) We want to test whether the score of **two classes** is adequately described by a normal distribution using $k = 5$ cells. What is your conclusion using **Chi-square goodness-of-fit test** with $\alpha = 0.05$? Please give your reasoning.

(b) Now we use 60 and 80 as dividing marks and obtain table 1. Are the scores and classes are independent with $\alpha = 0.05$? Write down your hypothesis, test statistic and your conclusion.

	$x < 60$	$60 \leq x < 80$	$x \geq 80$
Class I	8	5	2
Class II	4	7	4

Table 1: Dividing

(c) There are two more groups of scores, see table 2. Does the conclusion in (b) still hold? Please give your reasoning.

	$x < 60$	$60 \leq x < 80$	$x \geq 80$
Class I	8	5	2
Class II	4	7	4
Class III	10	15	5
Class IV	20	30	20

Table 2: Dividing

Solution:

(a) H_0 : The form of the distribution is normal H_1 : The form of the distribution is not normal. See the table

Class Interval	$x < 51.91$	$51.91 \leq x < 60.99$	$60.99 \leq x < 68.81$	$68.81 \leq x < 77.89$	$x \geq 77.89$
Observed Frequency O_i	7	5	6	4	8
Expected Frequency E_i	6	6	6	6	6

The test statistics is

$$\chi^2 = \frac{(6-7)^2}{6} + \frac{(5-6)^2}{6} + \frac{(6-6)^2}{6} + \frac{(4-6)^2}{6} + \frac{(8-6)^2}{6} = 1.67.$$

Since the two parameters in the normal distribution have been estimated, the chi-square statistic has $(5-2-1) = 2$ degrees of freedom. Since $\chi^2 = 1.67 < 5.9914 = \chi^2_{0.05}(2)$, we fail to reject H_0.

(b) H_0 : the two are independent; H_1 : the two are not independent. The test statistic is

$$\begin{aligned} T_2 &= \frac{(8 - 0.5 \times 0.4 \times 30)^2}{6} + \frac{(5 - 30 \times 0.5 \times 0.4)^2}{6} + \frac{(4 - 30 \times 0.5 \times 0.2)^2}{3} \\ &\quad + \frac{(4 - 0.5 \times 0.4 \times 30)^2}{6} + \frac{(7 - 30 \times 0.5 \times 0.4)^2}{6} + \frac{(4 - 30 \times 0.5 \times 0.2)^2}{3} \\ &= \frac{4}{6} + \frac{1}{6} + \frac{1}{3} + \frac{4}{6} + \frac{1}{6} + \frac{1}{3} \\ &= \frac{7}{3} = 2.33 \end{aligned}$$

Since $2.33 < 5.9914 = \chi^2_{0.05}(2)$, we fail to reject H_0.

(c) H_0 : the two are independent; H_1 : the two are not independent. We have table 3

	$x < 60$	$60 \leq x < 80$	$x \geq 80$
Class I	8(4.8461)	5 (6.5769)	2 (3.5769)
Class II	4(4.8461)	7 (6.5769)	4 (3.5769)
Class III	10 (9.6923)	15 (13.1538)	5 (7.1538)
Class IV	20 (22.6154)	30 (30.6923)	20 (16.6923)

Table 3: Dividing

The test statistic is $T = 5.2417 < 12.5916 = \chi^2_{0.05}(6)$, therefore we fail to reject H_0. The conclusion in (b) still holds.

4. We would like to investigate whether there is a relationship between weather conditions and the incidence of violent crime. We classified 1361 homocides according to the season, resulting in the following data: Test the null hypothesis of equal proportions using a significance level of $\alpha = 0.01$. Solution:

Winter	Spring	Summer	Fall
340	395	358	268

The chi-square test statistic is $\frac{(340340.25)^2}{340.25} + \frac{(395340.25)^2}{340.25} + \frac{(358340.25)^2}{340.25} + \frac{(268340.25)^2}{340.25} = 25.0779 > 11.34 = \chi^2_{0.01}(3)$, so the null hypothesis is rejected at 1%.

5. A bulb factory produced four batches of bulbs using filaments made of four different ingredients. A number of bulbs were randomly selected from each batch and their working life are measured as follows:

	1	2	3	4	5	6	7	8
A	1600	1610	1650	1680	1700	1720	1800	
B	1580	1640	1640	1700	1750			
C	1460	1550	1600	1640	1660	1740	1820	1820
D	1510	1520	1530	1570	1600	1680		

Please test whether the working life of those four batches of bulbs have significant difference($\alpha = 0.05$)

Solution:

Let $\eta_1, \eta_2, \eta_3, \eta_4$ be respectively the working life of the four batches of bulbs, which are four populations. $\eta_{i1}, \eta_{i2}, \eta_{i3}, \eta_{i4}$ are the samples of η_i, where we assume that $\eta_i \sim \mathcal{N}(a_i, \sigma^2), i = 1, 2, 3, 4$. Then we transform the question as to judge whether the null hypothesis $H_0 : a_1 = a_2 = a_3 = a_4$ holds. As

$$Q = \sum_{i=1}^{r}\sum_{j=1}^{n_i}\eta_{ij}^2 - \frac{1}{n}(\sum_{i=1}^{r}\sum_{j=1}^{n_i}\eta_{ij})^2$$

$$Q_A = \sum_{i=1}^{r}\frac{1}{n_i}(\sum_{j=1}^{n_i}\eta_{ij})^2 - \frac{1}{n}(\sum_{i=1}^{r}\sum_{j=1}^{n_i}\eta_{ij})^2$$

$$Q_e = Q - Q_A = \sum_{i=1}^{r}\sum_{j=1}^{n_i} n_i\eta_{ij}^2 - \sum_{i=1}^{r}\frac{1}{n_i}(\sum_{j=1}^{n_i}\eta_{ij})^2$$

And we denote:

$$R = \sum_{i=1}^{r}\sum_{j=1}^{n_i}\eta_{ij}^2,\ G = \sum_{i=1}^{r}\sum_{j=1}^{n_i}\eta_{ij},\ \eta_{i\cdot} = \sum_{j=1}^{n_i}\eta_{ij}$$

$$P = \sum_{i=1}^{r}\frac{1}{n_i}(\eta_{i\cdot})^2$$

Then we have

$$Q = R - \frac{G^2}{n},\ Q_A = P - \frac{G^2}{n}$$

$$Q_e = R - P$$

By calculation, we get the following table:

	n_i	$\eta_i^2\cdot$	$\eta_i^2\cdot/n_i$	G	P
A	7	560	44800	970	80549.17
B	5	310	19220		
C	8	290	10512.5		
D	6	-190	6016.67		

From the table we can get the $\frac{G^2}{n} = 36188.46$, and since $R = 231900$, we can get the following table by using the one-factor ANOVA table (where $n = 26, r = 4, n_1 = 7, n_2 = 5, n_3 = 8, n_4 = 6$) By checking

Source	Sum of Square(SS)	Degrees of Freedom	Mean Sqaure(MS)	F
Factor	44560.7	3	14786.9	2.15
Error	151351.3	22	6879.6	
Total	195712	25		

quantile table, we know $F_{0.95}(3, 22) = 3.05 > F = 2.15$, So we believe H_0 holds under the significant level of 0.95, which means the working life of these four batches of bulbs has no significant difference.

6. Data are given for the melting points for 50 metal alloy filaments. Here the data are repeated:

320	326	325	318	322	320	329	317	316	331
320	320	317	329	316	308	321	319	322	335
318	313	327	314	329	323	327	323	324	314
308	305	328	330	322	310	324	314	312	318
313	320	324	311	317	325	328	319	310	324

Test the hypothesis that these are observations of a normally distributed random variable with significant level 0.05. Note that you must estimate two parameters μ and σ.(Hint: using 10 categories)

Category	Class	observed	Expected	Contribution to Q
1	$(-\infty, 311.4465\}$	6	5	0.2
2	$(311.4465, 314.4165\}$	6	5	0.2
3	$(314.4165, 316.563]$	2	5	1.8
4	$(316.563, 318.3923]$	6	5	0.2
5	$(318.3923, 320.1]$	7	5	0.8
6	$(320.1, 321.8078]$	1	5	3.2
7	$(321.8078, 323.637]$	5	5	0
8	$(323.637, 325.7835]$	6	5	0.2
9	$(325.7835, 328.7535]$	5	5	0
10	$(328.7535, +\infty)$	6	5	0.2
				$Q = 6.8$

Since

$$\chi^2_{0.05}(7) = 14.067 > 6.8$$

The null hypothesis will not be rejected at 0.05.

7. (Microsoft stock price) Suppose that we have the daily stock price of a stock, say $S_1, S_2, \ldots, S_n$. We can then compute the so-called log-return of the stock price $R_1, R_2, \ldots, R_{n-1}$, where

$$R_i = \ln\left(\frac{S_{i+1}}{S_i}\right), \quad i = 1, \ldots, n-1.$$

Denote the median of R_i by m. On Blackboard, there is a dataset "MSFT_stock.csv" that contains the daily stock price of Microsoft from March 13 1986 to October 12 2020, with $n = 8718$. Use your favourite computing language to answer this question, and attach your code at the end of your assignment. **Do not use any existing packages or functions in your computing language that automatically conduct the tests for you. Instead, you can use any other functions or packages to help you conduct these tests.**

(a) We conduct the chi-square goodness-of-fit test, with the following H_0:

$$H_0 : R_i \overset{i.i.d.}{\sim} N(0, 0.02^2).$$

Fill in the following table: Do we reject the null hypothesis at $\alpha = 5\%$? What is the degree of freedom that you use?

Categories	Observed	Expected
< -0.001		
$[-0.001, -0.0004)$		
$[-0.0004, 0)$		
$[0, 0.0004)$		
$[0.0004, 0.001)$		
≥ 0.001		

(b) Conduct the chi-square goodness-of-fit test, with the following H_0:

$$H_0 : R_i \text{ is normally distributed.}$$

Estimate the two parameters by their maximum likelihood estimates and fill in the following table: Do we reject the null hypothesis at $\alpha = 5\%$? What is the degree of freedom that you use?

Categories	Observed	Expected
< -0.001		
$[-0.001, -0.0004)$		
$[-0.0004, 0)$		
$[0, 0.0004)$		
$[0.0004, 0.001)$		
≥ 0.001		

Solution:

(a) The test statistic is

$$2481.58 > \chi^2_{0.05}(5) = 11.0705,$$

so we reject H_0 at 5%. The degree of freedom is $6 - 1 = 5$.

Categories	Observed	Expected
< -0.001	3634	4184.69
$[-0.001, -0.0004)$	333	104.26
$[-0.0004, 0)$	151	69.55
$[0, 0.0004)$	377	69.55
$[0.0004, 0.001)$	317	104.26
≥ 0.001	3905	4184.69

(b) The test statistic is

$$475.81 > \chi^2_{0.05}(3) = 7.8147,$$

so we reject H_0 at 5%. The degree of freedom is $6 - 1 - 2 = 3$.

Categories	Observed	Expected
< -0.001	3634	3847.37
$[-0.001, -0.0004)$	333	220.69
$[-0.0004, 0)$	151	147.84
$[0, 0.0004)$	377	148.10
$[0.0004, 0.001)$	317	222.04
≥ 0.001	3905	4130.96